

Hyemin Doo

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Software Engineer

Full stack & AI-Focused Computer Engineering New Graduate with hands-on experience in data analysis, machine learning, and AI-powered systems. Strong interest in responsible AI, data lifecycle management, and risk-aware deployment of AI systems in regulated environments. Experienced in collaborating across engineering, analytics, and operational teams to turn complex data into trusted, actionable insights.

PROFESSIONAL EXPERIENCE

MENSUS (Early-stage semiconductor monitoring startup)

Jul 2025 - Dec 2025

Software Engineer

Seoul, South Korea

- Owned and shipped a **production full-stack monitoring system** for semiconductor equipment - actively used by Samsung Semiconductor's research team on the factory floor
- Built **React.js / TypeScript dashboard** (real-time charts, alert system, layout) and **Node.js / WebSocket backend** processing **>1kHz data streams**, improving monitoring responsiveness by **10–13%**.
- Engineered **WebSocket reconnect and heartbeat logic** - reduced connection drop frequency by **43%** and stabilized cross-platform communication between Windows and Linux systems.
- Deployed **MCP server with Claude Code CLI** for automated build/deploy workflows - saved **2–3 hours/day**; integrated Claude x Copilot validation pipeline saving **1+ hour per PR**.

Blue Ribbon Edu

Jun 2024 - Aug 2024

Data Analyst

Gangnam-gu, Seoul, Republic of Korea

- Built a **Python automation pipeline** for Excel data extraction, validation, and formatted delivery via KakaoTalk Messenger API - cut instructor manual review by **60%** across 200+ client records
- Led a 12-person squad through high-stakes mock missions - responsible for accurate data reporting, operational coordination, and team accountability under pressure

Republic of Korea Army

Aug 2022 - Feb 2024

Radar Operator (Squad leader)

Namhae-gun, Gyeongsangnam-do, Republic of Korea

- Built an internal records management system that cut information retrieval from **30–40 minutes to under 10 seconds**, eliminating repeated manual file checks across the unit.
- Led and coordinated a 12-person team during mock missions, ensuring **accurate data reporting, disciplined execution, and accountability** under high-pressure operational conditions.

PROJECTS

Inkling - AI Writing Platform | inklingwrite.com

Mar 2026 - Present

- Shipped a **production full-stack AI platform** (inklingwrite.com) - Next.js 15, TypeScript, Prisma/PostgreSQL (Supabase), Anthropic Claude API (Sonnet 4.6 + Haiku 4.5), PWA with push notifications and offline support.
- Built **RAG-powered essay generation** grounded in user's own writing, **Writing DNA** voice profile system (0–100% maturity score), and AI cost management with per-user spend caps and full admin observability.
- Engineered a **token-efficient model selection pipeline** - routing by task complexity across Sonnet 4.6 and Haiku 4.5 - reducing per-session AI cost by **93%** (\$0.30 → \$0.02 per user interaction).

Canvas AI - Full Stack AI Academic Assistant

Feb 2026 - Present

https://github.com/dannydoo123/canvas_ai

- Built a **full-stack AI study platform** that integrates with Canvas LMS - shipped both a React/Typescript web app and a Chrome Manifest V3 extension with injected content scripts and a background service worker.
- Implemented a **hybrid RAG pipeline** combining pgvector cosine search, BM25 keyword scoring, and **Hypothetical Document Embeddings (HyDE)** fused via Reciprocal Rank Fusion (RRF)
- Hardened **defense-in-depth security** including JWT auth with HTTP-only cookies and refresh token rotation, AES-256-GCM encryption for third-party credentials at rest, bcrypt password hashing, Zod input validation with private IP blocking, Helmet.js CSP headers, and endpoint-level rate limiting.

Comparing LLM Architectures in Street Fighter III

Sep 2025 - May 2026

Santa Clara University

- Built a browser-based platform to **benchmark LLMs** in real-time Street Fighter III matches, supporting both human-vs-model and model-vs-model sessions via WebRTC video streaming and a live MAME emulator backend deployed on Modal
- Benchmarked 5 LLMs/VLMs (**Qwen 3.5, Qwen 3.6, Gemma 4, Ministral 3, Nemotron 3 Nano**) under real-time constraints using a live Street Fighter III environment, exposing latency/cost trade-offs that offline leader boards obscure
- Built **dual inference pipelines** for multimodal models (JPEG frame + structured state via SGLang/vLLM) and a text-only path using a **custom YOLO detector** (ONNX + TensorRT) to extract spatial features for prompt construction, with SGLang constrained decoding via regex to enforce valid move outputs

TECHNICAL SKILLS

- AI tools & workflow:** Claude Code (CLI), Cursor, Codex (CLI), Prompt Engineering, MCP Server / Tool integration, LLM API integration, Retrieval Augmented Generation, Multi-Agent System Design
- Technical Skills:** Python, C/C++, HTML5, JavaScript, CSS, TypeScript, MongoDB, React, Node.js, Next.js, Machine Learning, PostgreSQL
- New Skill Group 1:**

EDUCATION

SANTA CLARA UNIVERSITY

Jun 2026

Bachelor of Science, Computer Engineering

- **Coursework:** Object-Oriented Programming and Advanced Data Structures, Abstract Data types and Data Structures, Embedded Systems, Algorithms, Web development, Artificial Intelligence (AI), CPU Architecture